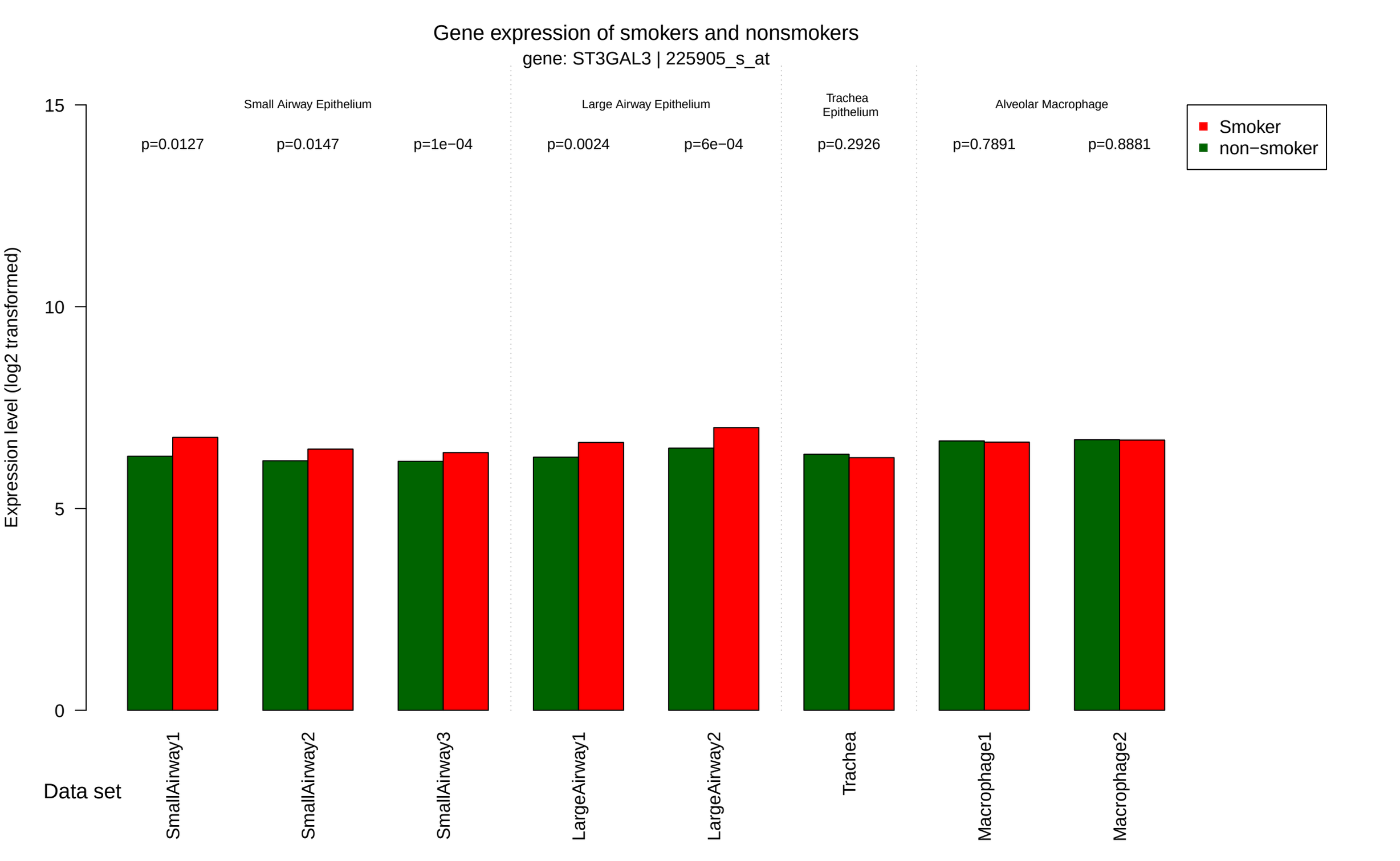
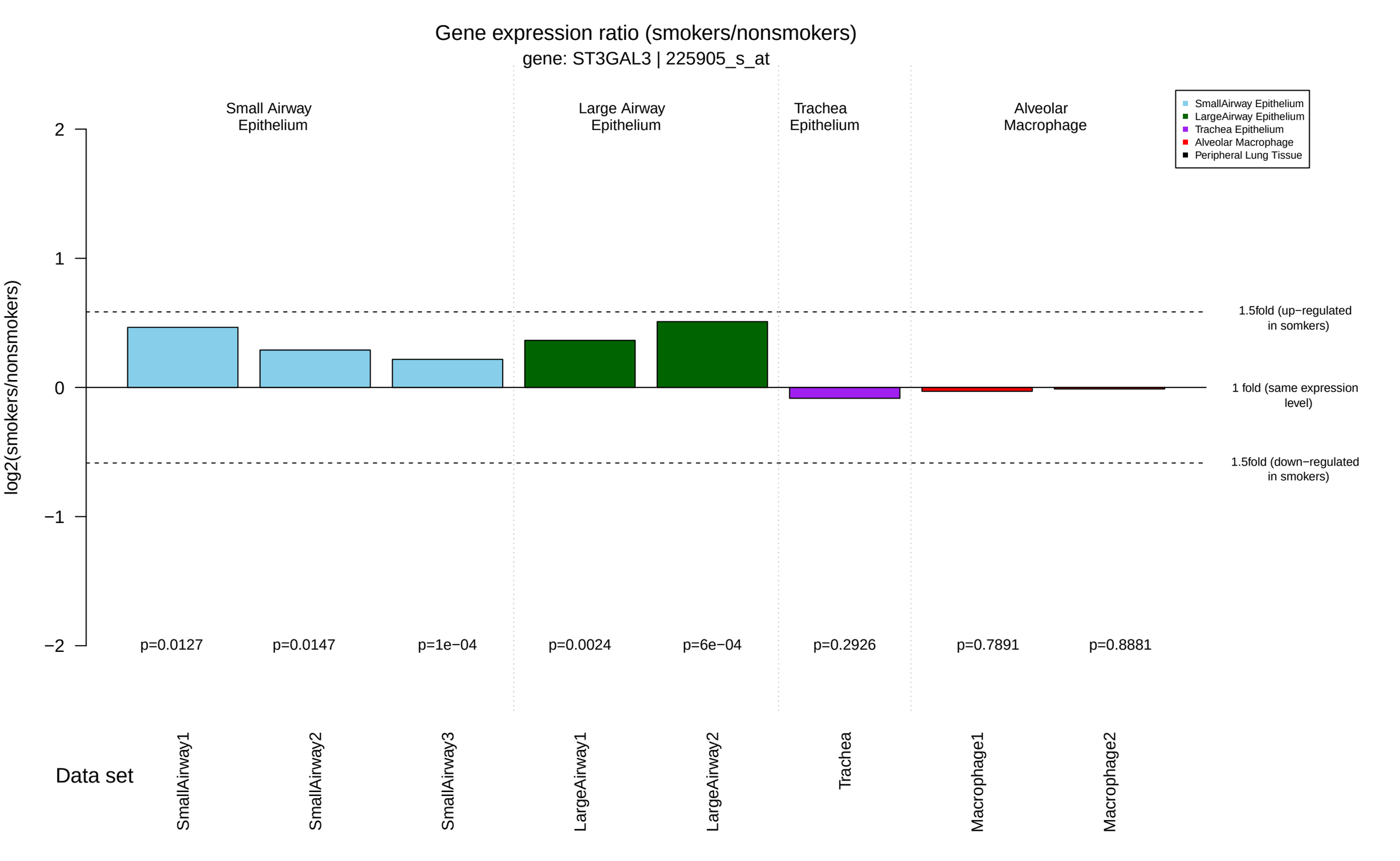
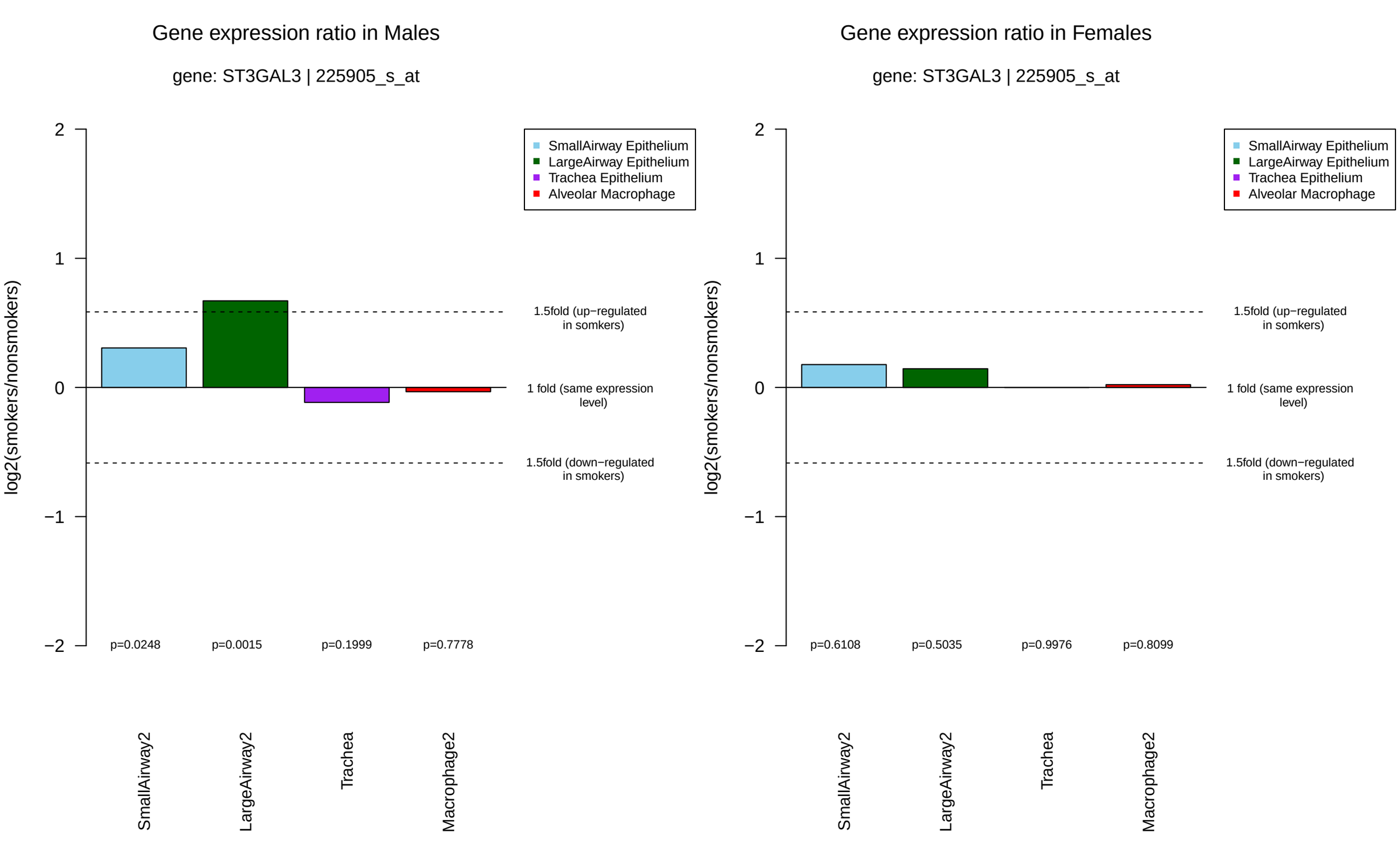
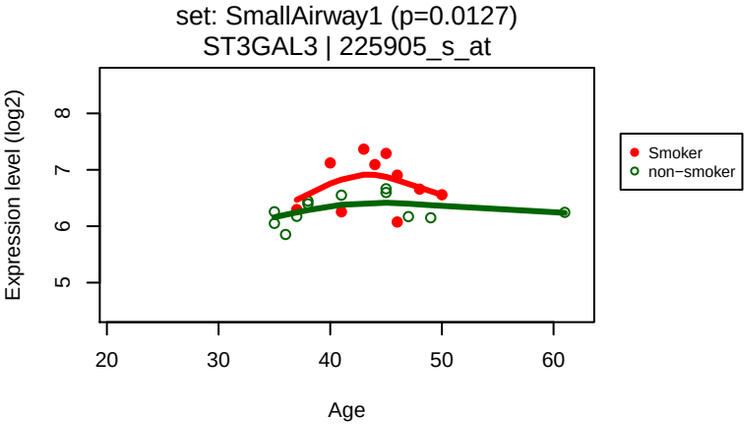
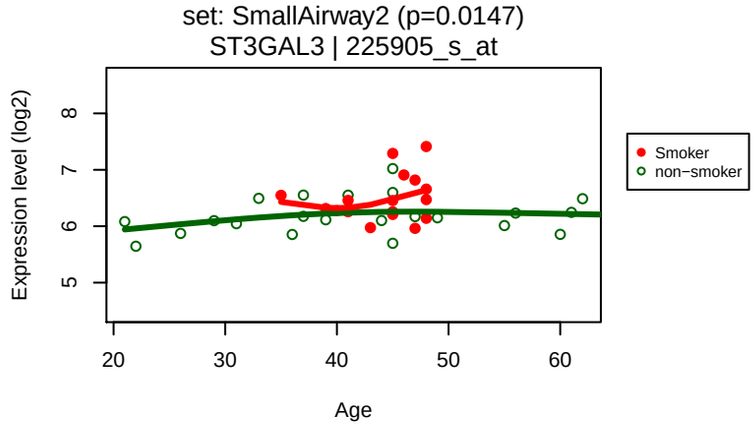
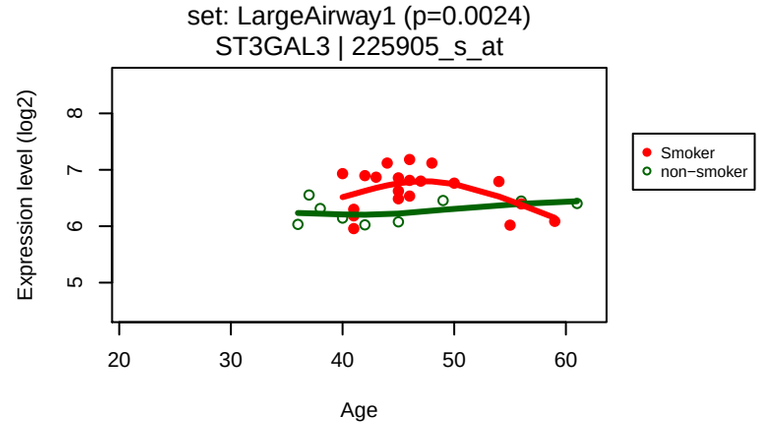

Gene expression of smokers and nonsmokers
gene: ST3GAL3 | 225905_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0127
p=0.0147
p=1e−04
p=0.0024
p=6e−04
p=0.2926
p=0.7891
p=0.8881
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

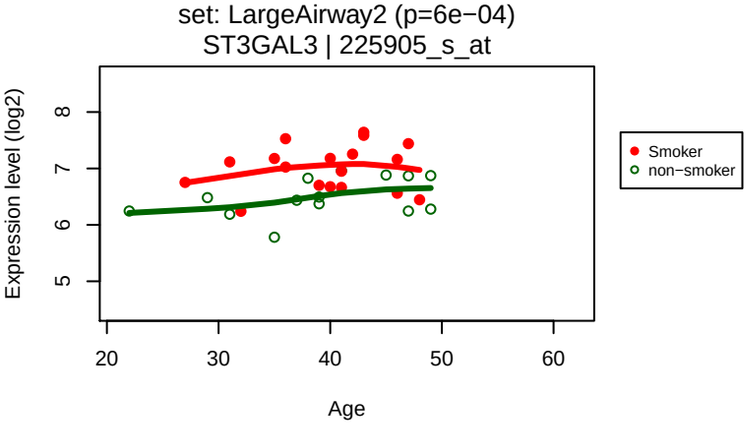

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL3 | 225905_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0127
p=0.0147
p=1e−04
p=0.0024
p=6e−04
p=0.2926
p=0.7891
p=0.8881
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

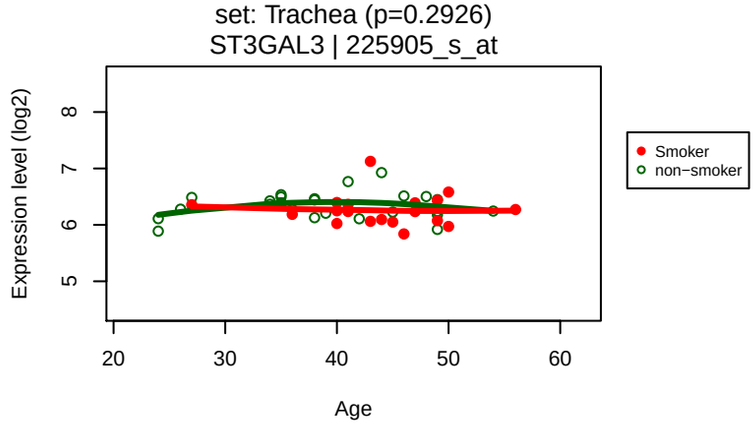

Gene expression ratio in Males
gene: ST3GAL3 | 225905_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0248
p=0.0015
p=0.1999
p=0.7778
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL3 | 225905_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6108
p=0.5035
p=0.9976
p=0.8099
SmallAirway2
LargeAirway2
Trachea
Macrophage2

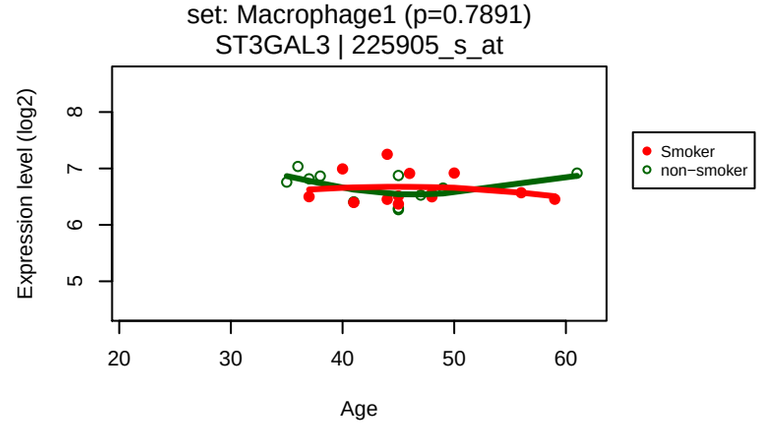

set: SmallAirway1 (p=0.0127)
ST3GAL3 | 225905_s_at
Expression level (log2)
Age
Smoker
non−smoker

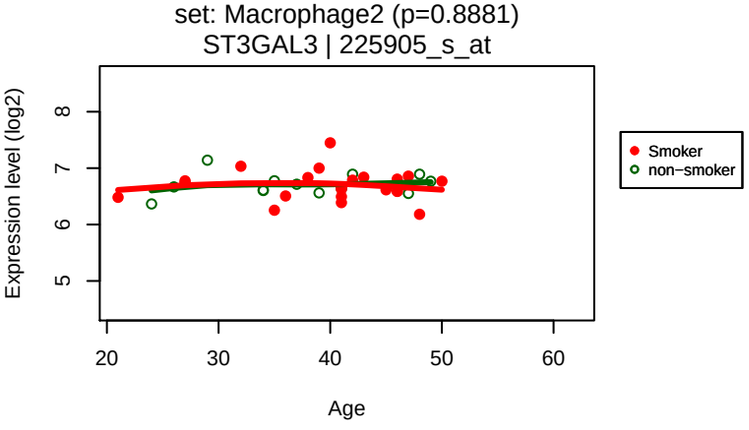

set: SmallAirway2 (p=0.0147)
ST3GAL3 | 225905_s_at
Expression level (log2)
Age
Smoker
non−smoker

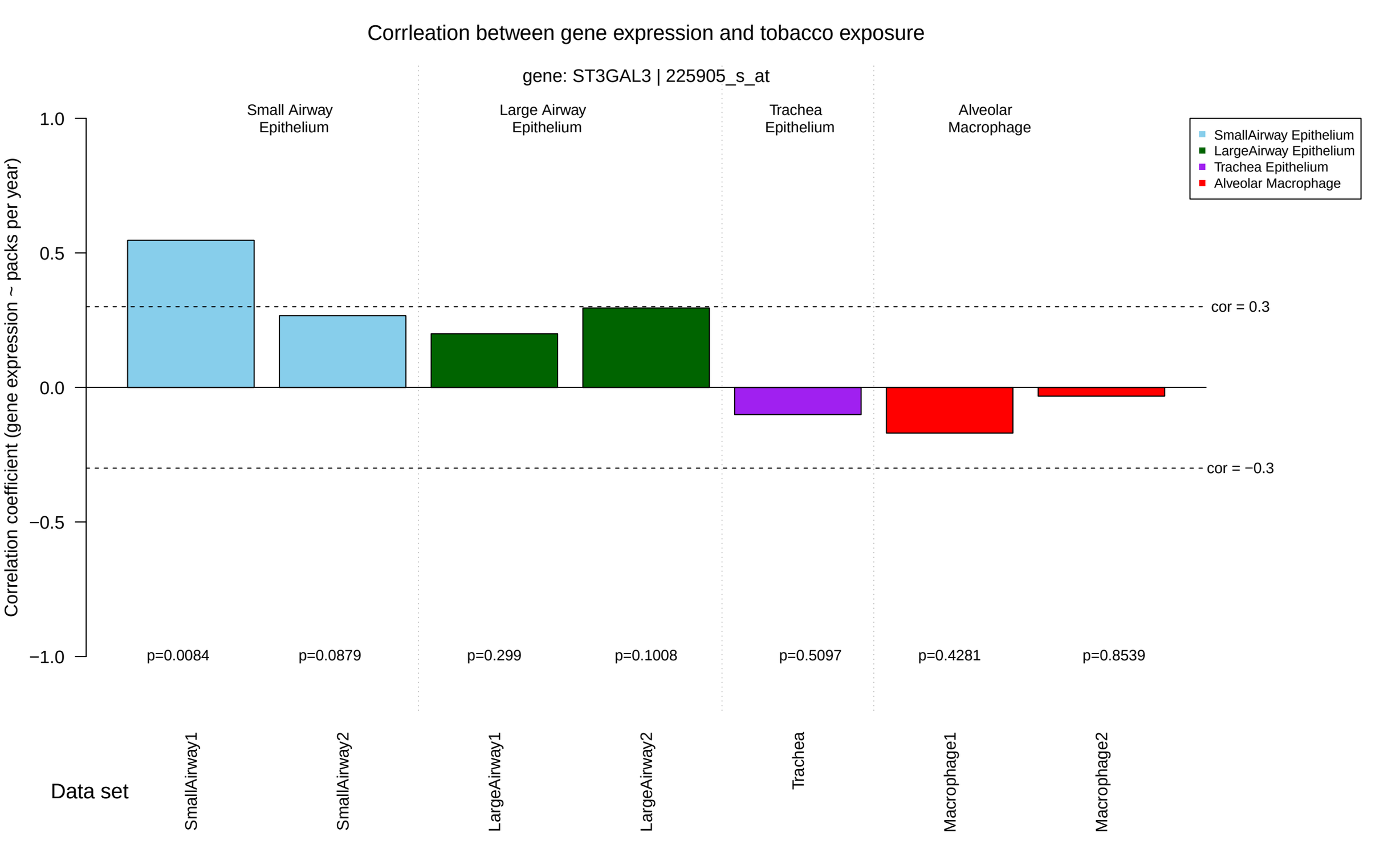

set: LargeAirway1 (p=0.0024)
ST3GAL3 | 225905_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=6e−04)
ST3GAL3 | 225905_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2926)
ST3GAL3 | 225905_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7891)
ST3GAL3 | 225905_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8881)
ST3GAL3 | 225905_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL3 | 225905_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0084
p=0.0879
p=0.299
p=0.1008
p=0.5097
p=0.4281
p=0.8539
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2